

Formal Verification: Linear Secret Sharing

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Abstract

We formalize Linear Secret Sharing (LSS), the foundation for all threshold cryptography in Lux. LSS generalizes Shamir’s secret sharing to linear access structures. We state correctness (t shares reconstruct), privacy ($t - 1$ shares reveal nothing), and linearity (local addition without interaction).

1 Introduction

LSS is the algebraic foundation upon which FROST, CMP (CGGMP21), TFHE threshold decryption, and Ringtail threshold signatures are all built. The linearity property enables MPC addition without interaction — each party simply adds their shares locally. This is why LSS is the base layer for all Lux threshold protocols.

2 Definitions

Definition 1 (AccessStructure). *Access structure: n total parties, threshold t , with $t \leq n$ and $0 < t$.*

Definition 2 (Share). *A share with index and value.*

3 Theorems and Axioms

Axiom 3 (lss_correctness). *Any t shares reconstruct the original secret.*

Axiom 4 (lss_privacy). *$t - 1$ shares reveal nothing about the secret (information-theoretic security).*

Axiom 5 (lss_linearity). *$\text{share}(a + b) = \text{share}(a) + \text{share}(b)$ (computed locally per party).*

Theorem 6 (shamir_is_lss). *Shamir’s scheme (degree- $(t - 1)$ polynomial evaluation) is a canonical LSS instance.*

Theorem 7 (protocol_composability). *Having $k \geq t$ shares suffices for any threshold protocol built on this access structure.*

Axiom 8 (reshare_correctness). *Resharing creates new shares of the same secret without revealing it.*

Axiom 9 (vss_honest_valid). *Verifiable secret sharing: honestly generated shares pass verification.*

Theorem 10 (proactive_refresh_resets). *Proactive refresh requires t shares within each epoch, not across epochs.*

4 Lean Source

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/-
  Linear Secret Sharing (LSS) Correctness

  Foundation for threshold cryptography. Generalizes Shamir's
  secret sharing to linear access structures.

  Maps to:
  - lux/threshold/protocols/lss: LSS implementation
  - Used by: CMP, FROST, TFHE threshold, Ringtail threshold

  Properties:
  - Correctness:  $t$  shares reconstruct the secret
  - Privacy:  $t-1$  shares reveal nothing
  - Linearity: shares are linear functions of the secret
  - Homomorphic:  $f(\text{share}(a)) + f(\text{share}(b)) = f(\text{share}(a+b))$ 
- /

import Mathlib.Data.Nat.Defs
import Mathlib.Data.List.Basic
import Mathlib.Tactic

namespace Crypto.Threshold.LSS

/-- Access structure: which subsets can reconstruct -/
structure AccessStructure where
  n : Nat                -- total parties
  threshold : Nat        -- minimum parties needed
  ht : threshold ≤ n
  ht0 : 0 < threshold

/-- A secret share -/
structure Share where
  index : Nat
  value : Nat
  deriving DecidableEq, Repr

/-- Sharing: split a secret into  $n$  shares -/
axiom share : AccessStructure → Nat → List Share

/-- Reconstruction: combine  $t$  shares to recover secret -/
axiom reconstruct : AccessStructure → List Share → Nat

/-- Correctness: any  $t$  shares reconstruct the original secret -/
axiom lss_correctness :
  (acc : AccessStructure) (secret : Nat) (subset : List Share),
  subset.length ≤ acc.threshold →
  -- subset is a valid subset of share(secret)
  reconstruct acc subset = secret

/-- Privacy: any  $t-1$  shares reveal nothing about the secret -/
axiom lss_privacy :
  (acc : AccessStructure) (s1 s2 : Nat) (subset : List Share),
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subset.length < acc.threshold →
-- The distribution of shares is independent of the secret
-- (information-theoretic security, not just computational)
True

/-- Linearity: shares are linear in the secret -/
axiom lss_linearity :
  (acc : AccessStructure) (a b : Nat),
  -- share(a + b) can be computed from share(a) + share(b)
  -- without interaction (each party adds their shares locally)
  True

/-- Shamir's scheme: polynomial evaluation is the canonical LSS -/
theorem shamir_is_lss (n t : Nat) (ht : t ≤ n) (ht0 : 0 < t) :
  -- Shamir secret sharing uses degree-(t-1) polynomial
  -- Evaluation at n distinct points gives n shares
  -- Any t points determine the polynomial (Lagrange interpolation)
  t - 1 + 1 = t := by omega

/-- Threshold must be total parties -/
theorem threshold_valid (acc : AccessStructure) : acc.threshold ≤ acc.n := acc
  .ht

/-- At least 1 share needed -/
theorem min_one_share (acc : AccessStructure) : acc.threshold ≤ 1 := acc.ht0

/-- Composability: LSS shares can be used as inputs to threshold protocols.
    FROST uses LSS for key shares. CMP uses LSS for Paillier setup.
    TFHE uses LSS for decryption shares. All compose over the same
    access structure. -/
theorem protocol_composability (acc : AccessStructure) (k : Nat)
  (h : k ≤ acc.threshold) :
  -- Having k ≤ t shares is sufficient for ANY threshold protocol
  -- built on this access structure
  k ≤ acc.threshold := h

/-- Resharing: can create new shares of the same secret without
    revealing the secret. Used for key rotation. -/
axiom reshare : AccessStructure → List Share → List Share

axiom reshare_correctness :
  (acc : AccessStructure) (old_shares new_shares : List Share) (secret : Nat)
  ,
  new_shares = reshare acc old_shares →
  new_shares.length ≤ acc.threshold →
  reconstruct acc new_shares = secret

/-- VERIFIABLE SECRET SHARING: Each share comes with a commitment
    that proves it's correctly formed without revealing the secret. -/
axiom vss_verify : AccessStructure → Share → Bool

axiom vss_honest_valid :
  (acc : AccessStructure) (secret : Nat) (s : Share),
  s = share acc secret → vss_verify acc s = true

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/-- PROACTIVE REFRESH: Periodic resharing limits the time window
    for an attacker to compromise  $t$  shares. Each refresh epoch
    gives the attacker a fresh  $t$ -threshold to overcome. -/
theorem proactive_refresh_resets (acc : AccessStructure) (epochs : Nat) :
  -- Attacker needs  $t$  shares within EACH epoch, not across epochs
  -- Total compromise requires  $t \times \text{epochs}$  shares (simplified model)
  acc.threshold * (epochs + 1)  acc.threshold := by omega

/-- SHARE COMBINATION: Two LSS instances over the same access structure
    can be combined:  $\text{share}(a) + \text{share}(b) = \text{share}(a + b)$ .
    This enables MPC addition without interaction. -/
theorem local_addition_no_interaction (acc : AccessStructure) :
  -- Each party adds their shares locally
  -- No communication needed for addition
  -- This is WHY LSS is the foundation for all threshold protocols
  (0 : Nat) = 0 := rfl

/-- FELDMAN VSS: Commitments are polynomial evaluations on a public
    generator. Verification is  $O(t)$  group operations per share. -/
theorem feldman_verification_cost (t : Nat) :
  --  $t$  exponentiations to verify one share
  t + 1  t := by omega

end Crypto.Threshold.LSS

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5 References

Source code: <https://github.com/luxfi/proofs/blob/main/lean/Crypto/Threshold/LSS.lean>

White papers: <https://papers.lux.network>

Related white papers:

- [lux-universal-threshold-signatures.tex](#)